

Trainings ▾

General Information

Oil technology advances due to requirements from engine technology (example: hotter combustion temperatures) or regulatory requirements (example: emissions regulations). There are multiple groups globally that provide standards for oil classifications. Among them are the following.

- American Petroleum Institute (API)
- Association des Constructeurs Européens d'Automobiles (ACEA)
- Japanese Automobile Manufacturers Association (JAMA)

The API oil category, or any international oil certification, is **not** sufficient by itself to select the appropriate engine oil for a Cummins® engine. Oils should be selected that meet the appropriate Cummins® Engineering Standard (CES). Table 1 below does provide a comparison between CES and other oil regulatory classifications for reference **only**.

Note : The CES number should not be considered to be sequential or supersede previous CES numbers unless specifically stated. Higher CES numbers do not necessarily indicate the oil is better than lower CES numbers.

The API requires a set of lab and engine tests that each commercial oil candidate **must** pass in order to meet the requirements for technology advances. API oil categories define the tests and properties required for a specific oil technology. Examples are below.

- CI-4 category oils were released in 2002 and updated in 2004, coinciding with exhaust gas recirculation technology, to improve soot control and shear stability.
- CJ-4 category oils were released in 2007, coinciding with emissions regulations and ultra low sulfur diesel fuel, to improve compatibility with aftertreatment, wear control, piston deposits, high temperature stability, and soot handling.
- CK-4 oils were released in December 2016 with all the typical viscosity grades of CJ-4 oils. As higher engine temperatures and lower emissions are a continuing trend, improved oils were required. CK-4 has improved oxidation resistance, shear stability, and aeration control. API category CK-4 is backward compatible with API category CJ-4 oil technology.
 - FA-4 oils were released in December 2016, primarily as a 10W-30 viscosity grade. FA-4 oils carry the same improved oxidation resistance, shear stability, and aeration control of CK-4 oils. FA-4 oils also reduce greenhouse gas emissions (increased fuel economy, dependent on duty cycle) by lowering high temperature/high shear (HTHS) viscosity. This lower HTHS results in directionally thinner oil films lubricating metal-metal interfaces. API category **FA-4 oils are not backward compatible** and should only be used when directed by the Lubricating Oil Recommendations and Specifications

procedure in Section V of corresponding owners/operation and maintenance manual to use CES 20087 engine oils.

Other global regulatory groups such as the ACEA and JAMA may require additional or different tests.

Table 1: CES Comparison to Other Oil Regulatory Classifications

CES	Closest API Category	Closest International Classification
Not Approved	CG-4	ACEA E1
CES 14615	None	None
CES 20074	None	None
CES 20075	CF-4	ACEA E2 ACEA E3
CES 20071 CES 20076 CES 20077	CH-4	ACEA E5 JAMA DH-1
CES 20078 CES20088	CI-4	ACEA E7
CES 20081	CJ-4	ACEA E9 JAMA DH-2
CES 20085	None	None
CES 20086	CK-4	None
CES 20087	FA-4	None
CES 20092	None	None